AWS STUDY NOTES

AWS Certificate Manager

A complete reference guide for ACM — provisioning, deployment, validation & best practices

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1 Introduction

AWS Certificate Manager (ACM) is a fully managed service that simplifies handling SSL/TLS certificates within AWS environments. SSL/TLS certificates are essential for encrypting data transmitted between servers and clients, ensuring secure access to websites and applications.

ACM provides a centralized solution for managing these certificates — including provisioning, deploying, and renewing — removing the complex manual processes involved.

> **CORE VALUE PROPOSITION**
>
> ACM removes the manual process of purchasing, uploading, and renewing SSL/TLS certificates. It integrates directly with key AWS services so that certificate deployment is seamless and automated.

Certificate Types at a Glance

Type	Use Case	Cost	Auto-Renewal
Public Certificates	Public-facing websites (e.g., `www.example.com`). Trusted by browsers by default.	Free	Yes (if DNS-validated)
Private Certificates	Internal resources — database servers, internal APIs, IoT devices within a private network.	Paid	Yes (if integrated)

2 Key Terminology & Prerequisites

Before exploring ACM's features, understanding the following foundational concepts is essential.

1. Certificate Authority (CA)

A **Certificate Authority (CA)** is a trusted organization responsible for issuing digital certificates. When a CA issues a certificate it verifies the identity of the requester, ensuring they are authorized to represent the domain or entity in question. CAs act as the trusted third party within **Public Key Infrastructure (PKI)**.

2. SSL / TLS Protocols

SSL (Secure Sockets Layer) and **TLS (Transport Layer Security)** are cryptographic protocols that establish secure connections over the internet. Although TLS has replaced SSL, the term "SSL" is still commonly used as a generic reference for secure communication.

3. Encryption in Transit

Encryption in Transit refers to encrypting data as it travels across networks — from client to server or between servers. This protects sensitive data such as login credentials, payment information, or personal data from interception or tampering during transmission.

4. SSL/TLS Certificates

Digital files issued by a CA that authenticate a website's identity and enable encrypted connections. Certificates contain:

- Information about the certificate owner
- The certificate's validity period
- The public key used for encryption

5. Public Key Infrastructure (PKI)

PKI is a system of digital certificates, public and private keys, and trusted CAs that enable secure communication over the internet. It underpins SSL/TLS by providing a framework for issuing, managing, and revoking digital certificates. AWS ACM helps manage these certificates within specific environments.

6. Certificate Expiration & Renewal

SSL/TLS certificates come with expiration dates, typically ranging from 90 days to several years. An expired certificate can disrupt service availability — browsers and clients will warn users or block access to the site. In managed environments like AWS ACM, public certificates are automatically renewed before expiration.

3 Types of Certificates in ACM

Public Certificates (Standard)

PUBLIC CERTIFICATE SUMMARY

- **Scope:** Public internet-facing sites (e.g., `www.google.com`)
- **Cost:** Free
- **Renewal:** Fully managed / automatic by ACM (if DNS-validated and in use)
- **Restriction:** Can only be used with integrated AWS services (ALB, CloudFront, API Gateway)

IMPORTANT RESTRICTION

You **cannot export** public ACM certificates for use on a standalone EC2 instance directly. You must use a Load Balancer or export a private certificate.

Public certificates are issued by external, trusted CAs and are designed for securing publicly accessible resources. They are widely accepted by browsers and client applications.

- Securing websites and load-balanced applications
- API Gateways with external clients
- CloudFront distributions

Private Certificates (ACM Private CA)

Private certificates, issued through **AWS ACM Private Certificate Authority (PCA)**, are ideal for securing internal applications. They do not rely on a publicly trusted CA, keeping them internal to your organization and reducing exposure.

Use Case	Example
Internal web servers	Internal HR portal or data analysis dashboard
Private APIs	Microservices communicating inside a VPC
Database servers	RDS instances requiring encrypted connections
IoT devices	Smart home devices communicating with AWS IoT Core

4 Key Use Cases for AWS ACM

Securing Elastic Load Balancers (ELB)

AWS ACM integrates seamlessly with all Elastic Load Balancer types (Application, Network, and Classic). Attaching an ACM certificate to an ALB enables HTTPS connections to secure user data flowing through the load balancer.

EXAMPLE

E-commerce websites can leverage ALBs with ACM certificates to securely handle payment information and customer data.

Amazon CloudFront for Global Content Delivery

When delivering content across geographical regions with CloudFront, ACM certificates enable HTTPS for secure communication. ACM makes it easy to deploy certificates globally.

CRITICAL EXAM RULE — CLOUDFRONT REGION

To use an ACM certificate with Amazon CloudFront, you **must** request or import the certificate in **US East (N. Virginia) — `us-east-1`** . CloudFront is a global service and its certificate management is centralized in `us-east-1` .

NOTE — HTTP VALIDATION FOR CLOUDFRONT

ACM now supports HTTP validation for CloudFront distributions in addition to the standard DNS and email validation methods.

EXAMPLE

Media streaming services can use CloudFront with ACM certificates to securely stream content to viewers worldwide.

API Gateway for Secure API Access

With API Gateway, you can assign an ACM certificate to a custom domain, ensuring secure access to your APIs. Particularly important for APIs that handle sensitive information or connect with external clients.

EXAMPLE

A financial institution providing secure access to customer data for authorized partners through a REST API can use ACM certificates for encrypted connections.

IoT Devices with AWS IoT Core

ACM certificates can be applied to IoT Core, enabling secure communication between IoT devices and the AWS cloud. Secure device-to-cloud communication is essential for IoT applications transmitting sensitive or personal data.

EXAMPLE

A smart home security system that sends real-time video and data to the cloud could use IoT Core with ACM certificates to ensure privacy and security.

Internal Applications Using ACM Private CA

ACM Private CA is ideal for internal applications that require SSL/TLS encryption but don't need publicly trusted certificates.

EXAMPLE

An internal HR portal or internal dashboard for data analysis can be secured with private certificates from ACM, ensuring only authorized users can access these applications.

5 Integration & Regionality

> **REGIONAL SERVICE**
>
> ACM is **not a global service** — it is regional. You must request your certificate in the same region as the service you are using (with the exception of CloudFront, which requires `us-east-1`).

Integrated Services

You can deploy ACM certificates only to specific integrated services. Public ACM certificates generally cannot be installed directly on a standalone EC2 instance.

AWS Service	Notes
Elastic Load Balancing (ALB, NLB)	Most common use case. Cert applied to the listener.
Amazon CloudFront	Cert must be in `us-east-1`.
Amazon API Gateway	Applied to custom domains.
AWS Elastic Beanstalk	Manages the underlying load balancer.
AWS CloudFormation	Can provision and reference ACM certs as resources.
AWS Nitro Enclaves	Secure enclave workloads.

Key Certificate Concepts

Concept	Detail
Standard Format	X.509 v3 Certificates
Validity Period	Public ACM certificates are valid for **13 months (395 days)**. This period cannot be changed.
Private Key — Public Certs	You **cannot download** the private key. It is managed internally by AWS.
Private Key — Private Certs	If exported from AWS Private CA, you can access the private key.
Encryption at Rest	ACM uses a unique, AWS-managed KMS key (`alias aws/acm`) to encrypt private keys at rest.
Immutable Domains	You **cannot add/remove domains** from an existing certificate. You must request a new certificate with the updated domain list.
Deletion Rule	You **cannot delete** a certificate while it is in use by an AWS resource (e.g., attached to a Load Balancer). Disassociate it first.

6 Domain Validation Methods

Before issuing a public certificate, ACM must verify you own the domain. There are three methods:

Method	How It Works	Auto-Renewal	Recommended?
DNS Validation	ACM provides a CNAME record to add to your DNS (Route 53 or external). Validation persists as long as the CNAME record exists.	Yes — forever	Best Practice
Email Validation	Sends an email to registered domain contacts (WHOIS). Requires a human to click a link. WHOIS lookup support is being deprecated/limited.	No — manual	Not recommended
HTTP Validation	Only available for certificates issued specifically for Amazon CloudFront distributions.	Limited	Niche use only

BEST PRACTICE

Always use DNS Validation when possible. It enables fully automatic renewal without any human intervention, as long as the CNAME record remains in your DNS configuration.

7 Expiration Management & Pricing

Managing Certificate Expiration with AWS Config and EventBridge

Ensuring SSL/TLS certificates do not expire unnoticed is a key operational challenge. AWS Config and EventBridge together provide a proactive solution.

Using AWS Config to Track Certificates

AWS Config monitors ACM resources and tracks certificate expiration dates. You can configure rules that trigger when a certificate is nearing expiration (e.g., 30 days before).

Setting Up Automated Alerts with EventBridge

EventBridge can detect Config events related to certificate expiration and trigger automated responses:

- **EventBridge Rule Triggers**
  Detects that a certificate will expire in 30 days via AWS Config event.
- **Lambda Function Invoked**
  Sends a notification via Amazon SNS or attempts automatic renewal action.
- **Notification Delivered**
  Administrators receive alerts via email, SMS, or other SNS-connected channels.

> **MONITORING ENHANCEMENTS**
>
> CloudWatch metrics and events are available for certificate expiration, availability, and renewal action required. CloudTrail logs all ACM API calls for audit purposes.

Pricing

Certificate Type	Cost	Notes
Public SSL/TLS	Free	You pay only for the AWS resources you attach them to (e.g., the Load Balancer).
AWS Private CA — General Mode	~$400/month per CA	Standard private CA operation.
AWS Private CA — Short-Lived Mode	~$50/month per CA	For certificates with shorter lifespans.
Per-Certificate Fee (Private)	From $0.75/cert	Sliding scale based on volume.
Imported Certificates	Free	Free to import and manage within ACM.

8 Best Practices & Conclusion

Best Practices

Practice	Why It Matters
Leverage ACM's Automatic Renewal	Ensures applications (CloudFront, ELBs) continue running without manual intervention. Use DNS validation to enable fully automatic renewal.
Limit Access via IAM Policies	Use IAM to control who can view or modify ACM certificates, limiting potential security risks from unauthorized modifications.
Use Private Certificates for Internal Services	ACM Private CA keeps certificates within your AWS environment, reducing risks associated with public exposure and allowing stricter lifecycle policies.
Monitor with Config + Automate with EventBridge	For mission-critical applications, set up multiple notification channels (SNS, email, SMS) to ensure no expiration goes unnoticed.
Always Use DNS Validation	Enables fully automatic renewal — no human action needed — as long as the CNAME record exists in your DNS.

Conclusion

AWS Certificate Manager (ACM) offers a comprehensive solution for SSL/TLS certificate management across AWS resources. From public to private certificates, ACM supports a wide range of use cases — including load balancers, API gateways, CloudFront distributions, and IoT endpoints.

By automating certificate renewal and implementing proactive monitoring with AWS Config and EventBridge, you can maintain secure connections and eliminate the risks of expired certificates. This streamlined approach reduces operational overhead, allowing your team to focus on higher-value activities while ACM handles the complexity.

> **KEY TAKEAWAY**
>
> For any application that handles sensitive data or requires encrypted communication, **AWS ACM is an essential tool** in your AWS toolkit. Its automation, integration breadth, and cost model (free public certificates) make it the default choice for TLS management on AWS.